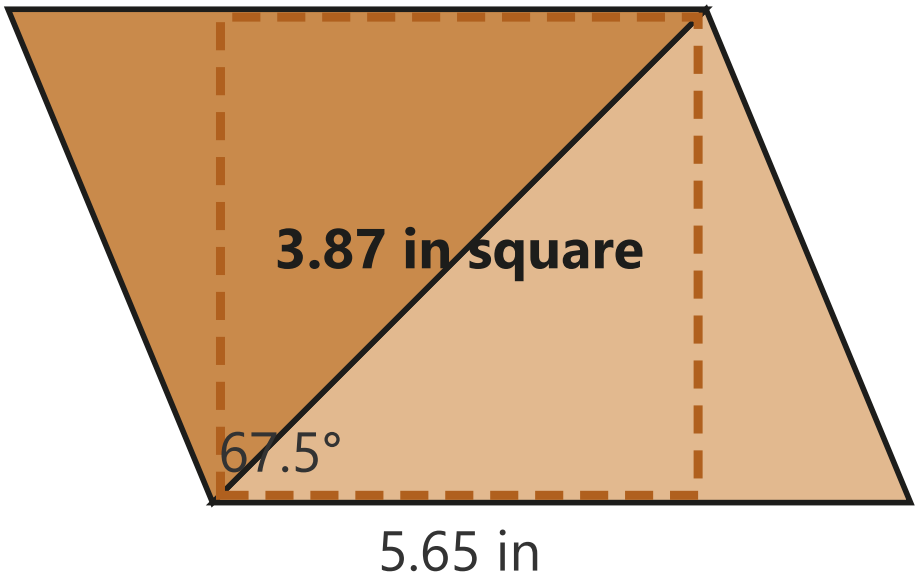
‘Square’ on Demand

Building with wedge-shaped lumber



DONOVAN ZEANAH

‘Square’ on Demand: Building with wedge-shaped lumber

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Built 14 October 2026 by the DomeSim project’s own software; every figure is computed from the geometry of a split log, and the few that are declared are listed at the back.

This book describes methods, not rated structures. Members made from paired wedges are not graded lumber. Anything that carries people, snow or wind is checked by a licensed engineer before it is built.

github.com/dkzeanah/DomeSimulator

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BEFORE YOU BEGIN

What This Book Is

A round log split through its centre does not give you boards. It gives you wedges: long sticks whose two flat faces meet at the pith, with the bark on their backs. The dome books in this project build a whole house out of them, because a triangulated frame does not care what shape its sticks are.

This book asks the next question. What else can you build with wedge-shaped lumber, and what do you do when a job really does want a square stick?

The answer is in the title. You do not have to choose between wedges and boards when you fell the tree. Keep the wood as wedges, which is how a log gives up the most of itself, and make a square member only where a square member is needed: two wedges put together the right way round, planed true, on demand. Everywhere else, use the wedge as it is, and use its angles, which turn out to be useful in their own right.

Everything here is worked from one log: the project's reference trunk, 12 inches across, split 8 ways. Every number is computed from the geometry, by the same software that solves the domes, so you can change the log and get a new book. The few figures that are not computed – the saw's kerf, the planing allowance, the standard sizes of dressed lumber – are declared in a table at the back, with where each one comes from.

WHO THIS IS FOR

Owner-builders with a chainsaw and a stand of timber. Veterans' build crews and the villages they serve. Teachers who want geometry that ends in something you can stand inside. And anybody who has watched a mill turn a good log into a pile of slabs and wondered whether there was another way.

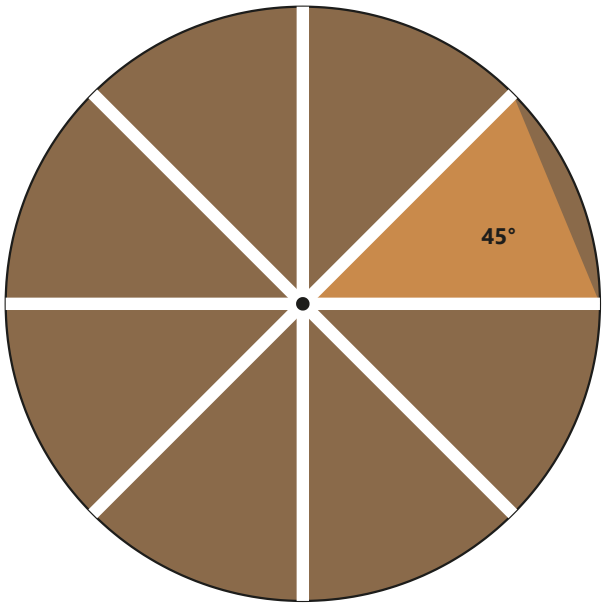
BEFORE ANYTHING ELSE

This is a book of methods, not engineering ratings. A member made from paired wedges is not graded lumber, and nothing in these pages tells you what load it will carry. Anything that holds up people, snow or wind is checked by a licensed engineer before it is built. The saw work is adult work, done in full protective gear, and never alone.

CHAPTER 1

Why Wedges

A log is round, and almost everything we build is square. The sawmill settles that argument in favour of the square: it saws rectangles out of the circle and sends the curved outside away as slabs, edgings and sawdust. Splitting settles it the other way. It accepts the circle and cuts it into sectors, so that nothing is taken off the outside at all.



A 12 inch log split 8 ways through the pith. Each cut runs straight through the centre, so every wedge is the same shape and nobody measures an angle.

The log does the geometry

Halve a circle through its centre, halve each half, halve each quarter, and you have eight sectors of exactly 45.0 degrees each, whether or not you own a protractor. The method needs a straight cut through the centre and nothing else. That is why a chainsaw and a steady hand can produce matching members by the hundred: the accuracy is in the log, not in the person.

Every face goes through the heart

Wood shrinks as it dries, and it shrinks more around the growth rings than across them: from green to oven-dry, 7.4 per cent around the rings against 4.8 per cent across them, for the pine these books use. A sawn board cut away from the centre has faces that cut the rings at every angle, so it cups, bows and twists as it dries. A wedge's two flat faces both run through the pith, square across the rings. It shrinks mostly in its width, evenly, and stays straight. Of all the ways to cut a log, the radial cut moves least.

A wedge carries its wood where it counts

A wedge is deep, pith to bark, and narrow at the point. Stood on its back, it puts most of its wood far from the middle, where a beam needs it. In a triangulated frame – a dome, a truss, a zome – the members are pushed and pulled along their length rather than bent, and the shape of the stick hardly matters. That is the whole argument of the dome books: *triangles do not care what shape your stick is*.

THE SHORT VERSION

A split keeps more of the log, moves less as it dries, needs no measured angle, and comes off one small chainsaw. The cost is that you get wedges, not boards. The rest of this book is about when that is a cost at all.

SEE IT FOR YOURSELF

Draw a circle on paper and cut it out. Fold it in half, in half again, and once more, and unfold it: eight matching sectors. Now draw the largest rectangle you can inside a second circle and cut it out. Lay the four curved scraps next to your eight sectors. Which way kept more of the paper?

CHAPTER 2

The Wedge, Measured

Before pairing wedges or shimming them, know the one you have.

A wedge from a log of radius R, split n ways, has:

- two flat faces, each R long, meeting at the pith at 45.0 degrees for an eighth;
- a curved back. Trimmed flat, the back is a chord 4.59 inches wide on the reference log, and the section becomes an isosceles triangle;
- a depth, pith to the middle of the back, of 5.54 inches;
- two corners at the back of 67.5 degrees each.

For the reference log that triangle is 12.7 square inches. The sector it came from was 14.1; trimming the back flat costs 1.4 square inches, 10 per cent of the wedge. Leave the back round where nothing has to sit on it, and you keep that wood too.

The saw takes its share

A chainsaw cut is not a line. The saw removes a strip 0.25 inches wide, and splitting 8 ways takes four cuts across the whole diameter. On the reference log that is 11 per cent of the round. Every wedge loses half a kerf off each flat face, which moves its point outward a little; the triangle that is left is the same shape, slightly smaller. Every figure from here on has the kerf taken out.

Bigger logs, bigger wedges

The proportions never change: an eighth is always 45.0 degrees at the point and 67.5 at the back. Only the size does.

log, in	wedge back, in	pith to back, in	pair planes to, in square	dressed 2x4s in it	dressed 4x4s in it
8	3.06	3.70	2.45	0	0
10	3.83	4.62	3.16	0	0
12	4.59	5.54	3.87	2	1
14	5.36	6.47	4.57	2	1
16	6.12	7.39	5.28	3	1
18	6.89	8.31	5.99	3	1

The last three columns are the subject of the next chapter: two wedges of each size, put together and planed square.

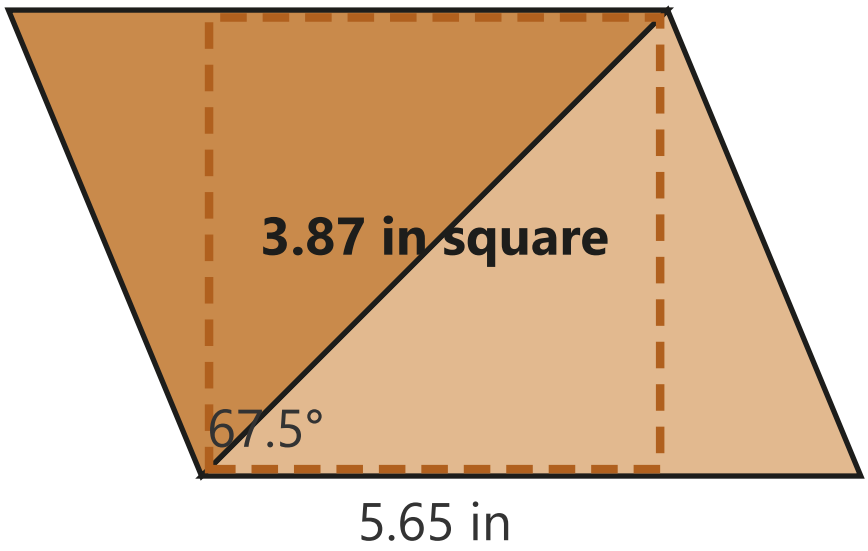
WHY THE ANGLES NEVER CHANGE

An eighth of a circle is an eighth of a circle at any size. The point angle is three hundred and sixty degrees divided by the number of splits; the back angles are what is left of a hundred and eighty, shared equally. Change the log and every length scales together, but the shape is fixed by how many ways you split it.

CHAPTER 3

Square on Demand

Take two wedges. Turn one end for end, so its point lies against the other's back, and lay them together on a flat face – slant to slant. What you are holding looks like a two-by-four that has been pushed sideways: four flat faces, two long and two short, with corners of 67.5 and 112.5 degrees instead of ninety.



Two eighths, slant to slant. The pair is a parallelogram, 5.65 by 4.32 inches with corners of 67.5 and 112.5 degrees. The dashed square is what it planes to.

The skew planes to a square

Here is the part that gives this book its name. The largest rectangle you can plane out of that skewed pair, cut square to its long faces, is $R \cos \theta$ wide and $R \sin \theta$ thick, where θ is the angle at the wedge's point. For an eighth, θ is 45.0 degrees, and the cosine and the sine of 45.0 degrees are the same number, 0.707. So the rectangle is a square.

On the reference log, after the kerf and a planing allowance of 1/16 inch a face, two eighths plane to a member **3.87 inches square**. That is enough for 1 dressed four-by-four, or 2 dressed two-by-fours resawn from it.

What it costs

Squaring is not free. The planed square keeps 66 per cent of the pair, and 53 per cent of the two sectors of round log it came from. Split eighths keep 89 per cent of that same log as they come off the saw. So a square is something you pay for out of the wedge's advantage, and you should buy it only where you need it: a post that has to take a door frame, a sill that sits on a level foundation, a plate a wall stands on.

That is the method in one line. **Split everything; square on demand.**

MAKE A SQUARE FROM TWO WEDGES

1. Choose two wedges from the same section of the same log, so they match.
2. Plane or flush-cut the backs flat if they are to be glued; leave the radial faces as sawn if they mate well.
3. Turn one end for end and lay them slant to slant. Clamp, and either glue the joint or bolt through it at a spacing your engineer gives you.
4. Mark the square on both ends with a carpenter's square, square to one long face, and plane down to the marks.

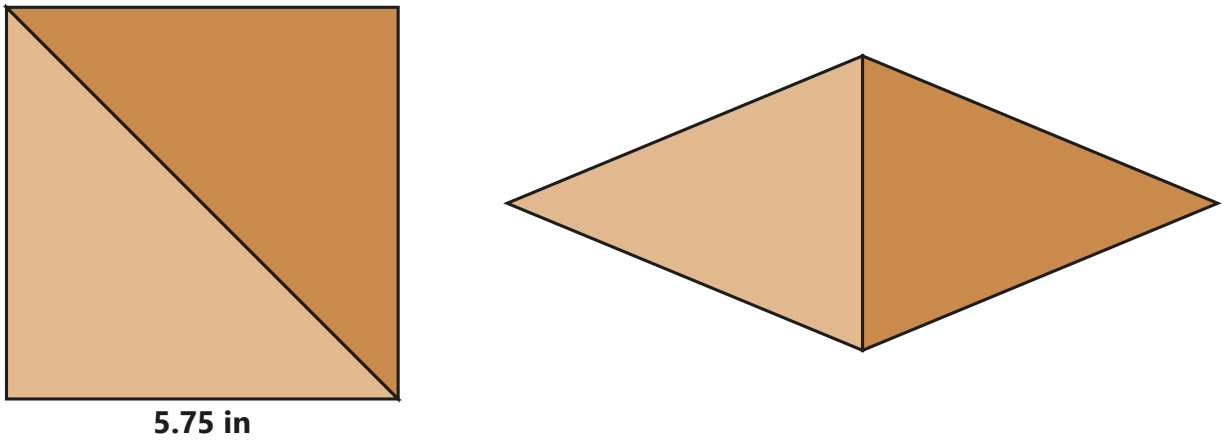
WATCH OUT

A glued pair is only as straight as its two wedges. Pair wedges from the same section, dry them together, and glue them dry. A bolted pair can be taken apart and used as wedges again; a glued one cannot.

CHAPTER 4

Back to Back

There is a second way to put two wedges together: on their backs, the trimmed bark faces glued or bolted face to face, points out.



Quarters back to back are a true square with nothing to plane. Eighths back to back make a diamond.

Quarters make a square

Split a log only four ways and each quarter is a right-angled triangle once its back is trimmed. Two right-angled triangles laid together on their long sides are a square. There is nothing to plane except the saw marks: two quarters of the reference log make a member **5.62 inches square**, planed, and keep 56 per cent of the round they came from.

That is the heaviest square the method makes, and it is the one for posts. A quarter is also the easiest split there is: two cuts across the log, at right angles, through the pith.

Eighths make a diamond

Two eighths back to back make a diamond – a rhombus whose sharp corners are 45.0 degrees. It is a poor source of boards: the largest board inside it is 3.87 by 1.53 inches, 21 per cent of the wood. But a diamond is the right shape for a different job: a ridge, a rail with both faces shedding water, a spline driven into a V-groove, or a corner post for an octagon.

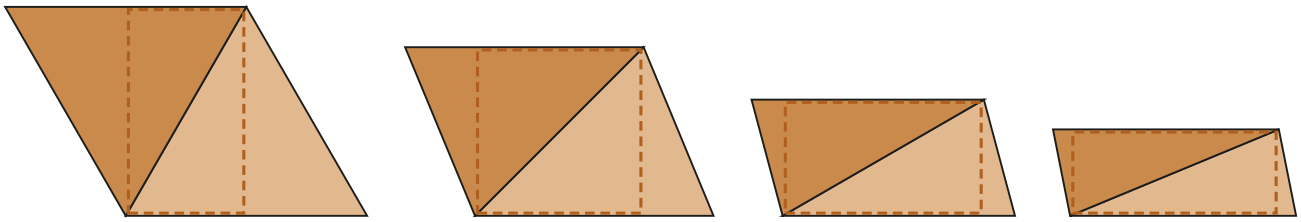
WHY THE QUARTERS COME OUT SQUARE

A quarter's point is a right angle, and its two flat faces are the same length. Put two of them together along the side opposite the point and you have four equal sides and four right angles. The eighth's point is half a right angle, so the same move gives four equal sides and two sharp corners: a diamond.

CHAPTER 5

Split Finer, Get Boards

The skewed pair keeps cos θ of itself when it is planed square to its long face. The smaller the wedge's point angle θ, the closer that is to all of it: a thin wedge pairs into something very close to a board. Split finer, and the pairs come out flatter, wider and less wasteful to plane.



Pairs from the reference log, split six, eight, twelve and sixteen ways, each with the board it planes to. The finer the split, the flatter the pair.

The saw pushes back

There is a price. Every extra split is another cut across the log, and every cut takes a kerf. Split 8 ways and the saw takes 11 per cent of the round; split twenty-four ways and it takes 32 per cent, and what is left keeps only 59 per cent as planed boards. Somewhere in between, the two effects balance.

split	point, deg	pair	member, in	members	kept %	kerf %	skew pitch, in 12
4 ways	90.0	back to back	5.62 x 5.62	2	56	5	12.0
6 ways	60.0	slant	4.82 x 2.73	3	35	8	6.9
8 ways	45.0	slant	3.87 x 3.87	4	53	11	5.0
12 ways	30.0	slant	4.64 x 2.63	6	65	16	3.2
16 ways	22.5	slant	4.81 x 1.92	8	65	21	2.4
24 ways	15.0	slant	4.74 x 1.18	12	59	32	1.6

On the reference log the balance falls at **16 ways**, where pairs plane to boards 4.81 by 1.92 inches and keep 65 per cent of the log. Twelve ways is nearly as good – 65 per cent, as members 4.64 by 2.63 inches – with fewer cuts.

Choosing your split

- **Four ways** for posts: quarters back to back, square without planing.
- **Eight ways** for frames: wedges as they are, and squares where needed.
- **Twelve or sixteen ways** for boards: pairs that plane to near-dimensional stock with the least waste.

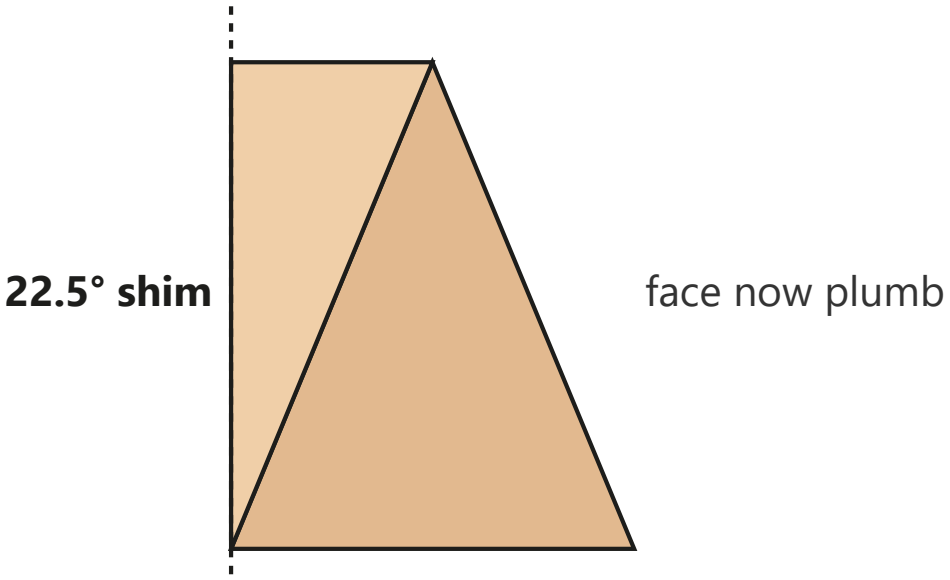
You do not have to split a whole log one way. Quarter the butt for posts, split the middle eight ways for frame members, and take the top sixteen ways for boards. The log tells you which: big and straight, square it; small, split it fine.

> **WATCH OUT**
>
> Splits that are not a power of two cannot be made by halving. Twelve ways means marking the end grain with a template and cutting to the marks, and a cut that wanders off the centre gives you wedges that will not pair. Mark both ends, and keep the bar on the line.

CHAPTER 6

Shims From the Same Log

Sometimes a wedge has to meet a square world for a short distance: where a strut lands on a level plate, where it passes a door jamb, where a triangular member turns into a square one. You do not have to square the whole stick for that. Shim it where it needs to be square, and leave it a wedge everywhere else.



An eighth lying on its back, with a thin wedge set against one face. The shim's angle is half the wedge's point, so the face it makes is plumb.

The shim is a thinner wedge

An eighth lying on its back presents two sloping faces, each at 67.5 degrees to the ground. To make one of them plumb you need to add 22.5 degrees – exactly half the wedge's point. And a wedge with a 22.5 degree point is what you get when you split a log 16 ways. **The shims come from the same log: halve one more time.** Over the full face of a reference wedge, that shim is 2.49 inches thick at its heel.

Where to shim

- **At a seat.** Where a strut bears on a sill or a plate, shim the last foot so the bearing is square and full-faced. The rest of the strut stays a wedge.

- **At a transition.** Where a triangulated frame meets a square-framed wall, a pair of shims either side turns a wedge into a stud for the length of the joint.
- **Along the axis, at points.** A long wedge can be shimmed at intervals – every place a cross member or a fixing lands – instead of being planed square for its whole length.
- **To take up the skew.** A skewed pair laid on a flat surface rocks on one long corner. A tapered shim under its low side makes it sit square without planing.

MAKE A SET OF SHIMS

Take a short, straight section from the top of the log. Split it 16 ways. Trim the backs flat. You now have a box of identical 22.5 degree tapers, matched to every wedge from that tree.

WHY HALF THE POINT

A wedge on its back has corners of 67.5 degrees at the ground. A plumb face needs ninety. The difference, ninety minus 67.5, is always half the point angle, whatever the split.

CHAPTER 7

Use the Skew

Planing the skew out is one choice. The other is to keep it, because buildings are full of angles that are not ninety degrees, and the wedge brings its own.

The skew is a roof pitch

A skewed pair laid flat has its short faces sloping at half the wedge's point angle – 22.5 degrees for an eighth. As a roof pitch that is a rise of **5.0 in twelve**: almost exactly the five-in-twelve of an ordinary roof. Use pairs of eighths as rafters or purlins and their faces already lie at the slope, so sheathing and roofing seat flat without a single bevel cut.

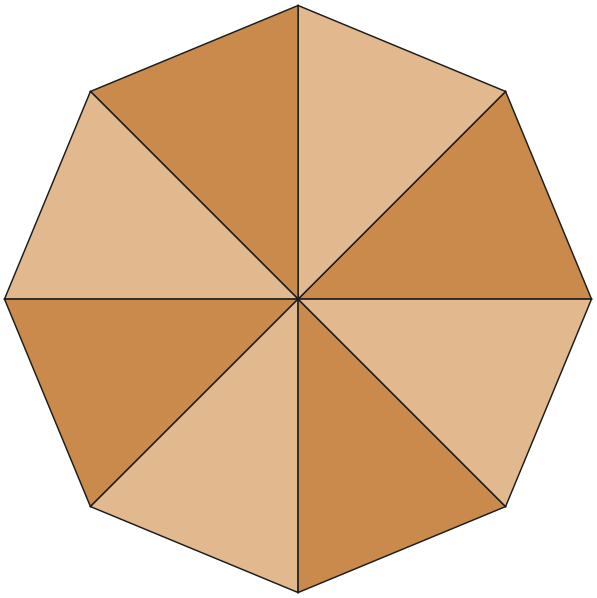
Other splits give other pitches:

- quarters: 12 in twelve, a steep forty-five degree roof;
- sixths: 6.9 in twelve;
- twelfths: 3.2 in twelve, a low shed roof;
- sixteenths: 2.4 in twelve.

Choose the split for the roof you want, and the wood arrives cut to the pitch.

The octagon is already in the log

An eighth's point is 45.0 degrees – the angle at the centre of an octagon. Eight trimmed wedges laid points-in close into an octagon with every joint the log's own angle, and every outside corner the 135 degrees an octagon needs.



Eight trimmed wedges, points in. An octagonal floor, table or gazebo deck is the log put back together, with its pieces flat.

That makes octagonal decks, gazebo floors, round tables, and the radial joists of a small round building into the most natural thing to build from wedges: the members meet at the centre without a single angle being measured.

Other places to keep the angle

- **Bevel siding.** A wedge sawn along its length into thin slices gives tapered boards that lap like clapboard, thick edge down.
- **Trusses and A-frames.** A triangulated truss has members meeting at angles; the wedge's faces seat against each other without a square edge in sight.
- **Sloped sills and drip edges.** A wedge laid on its back sheds water both ways.
- **Fences and rails.** A wedge rail on its back is a capping that does not hold rain.

> **PLANE IT, OR KEEP IT**
>
> Square where the building is square: posts, plates, door and window frames, anything that meets a level floor. Keep the skew where the building already slopes. Every face you do not plane is wood you keep and an hour you save.

CHAPTER 8

Joints Wedges Make Easy

Wedge lumber has flat faces at known angles, a point, and a back. Each of those is the start of a joint that square lumber has to be cut to make.

The pinwheel

The dome books' joint. Where several members meet at a corner, none of them runs into the corner: each stops a little short and bears on the side of the next, round the corner like the blades of a pinwheel. Nothing meets at a point, so nothing has to be cut to a compound angle there. It works with sticks of any section, which is why it suits wedges.

The slant lap

Two wedges laid slant to slant already lap along their whole length. Offset one by a foot and you have a scarf: a long member made of short wedges, the joints staggered so no two fall together. Bolt or peg through the lap.

The bird's mouth on the point

A wedge's point is a ready-made bird's mouth. Lay it point-down over a square plate and the plate's corner sits in the angle between the two faces. Cut the point back by a little and it seats on a square corner of any size.

The keyed seam

Two wedges side by side on their backs open a V between them. Drive a diamond or a thinner wedge into the V and it locks them together. Leave it open and it is a channel – which is the next chapter.

The butterfly and the dowel

A pair glued slant to slant can be stitched across its joint with a butterfly key let in across the seam, or held with dowels through both wedges. Neither needs a square face to reference from: the joint line is the reference.

Flush-cut in place

Wedges are not cut to finished length on the bench. Set them in the frame long, then run the saw across the joint so both pieces are cut by the same pass. Whatever the angle is, both faces get it, and they meet.

Jigs that hold a triangle

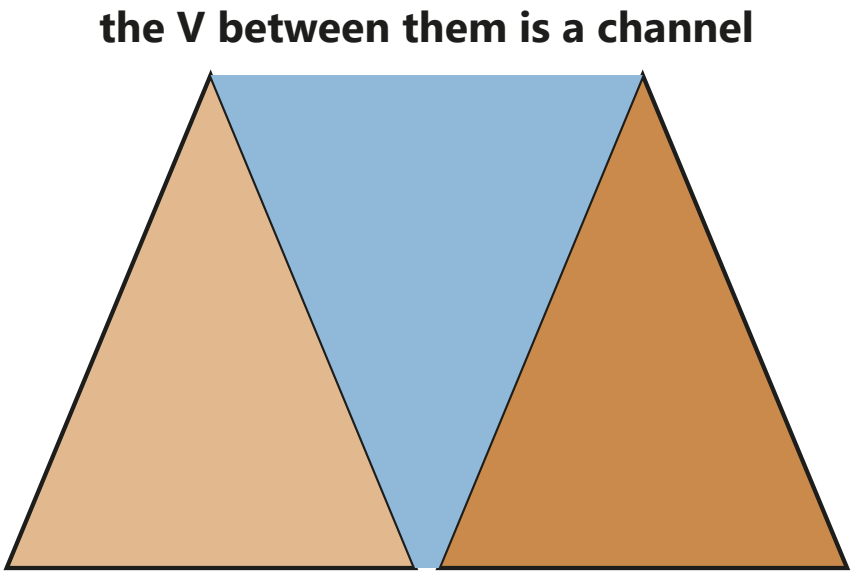
Every one of these joints is easier with a jig that holds a triangular section still: two boards screwed into a V at the wedge's point angle. A wedge dropped into it sits on its faces, back up, and cannot roll. One jig per split angle does every wedge from that split.

> **WATCH OUT**
>
> Fasteners near a wedge's point split it. Keep screws and bolts in the thick part, toward the back, and drill pilot holes in the thin part every time.

CHAPTER 9

The Seam Is a Channel

Square lumber laid side by side closes up. Wedges laid side by side on their backs do not: their sloping faces open a V between them, the full length of the member.



Two wedges on their backs, a kerf's width apart. The V between them runs the whole length, and it can be a channel instead of a gap.

In the dome books that gap is the most useful thing in the building. Capped outside, the seam between panels becomes a duct: it carries rain off the shell to the tank, wiring round the room, and air to wherever it is needed, without a single gutter or conduit added. The same is true of any wall or floor laid up from wedges.

What fits

- **Water.** A capped V seam is a gutter that cannot overflow sideways, sloped by the building itself.
- **Wire.** A seam is a cable run that is never buried in a wall: lift the cap and the wiring is there.
- **Air.** A seam can breathe a wall, carrying damp air out of it.

What does not

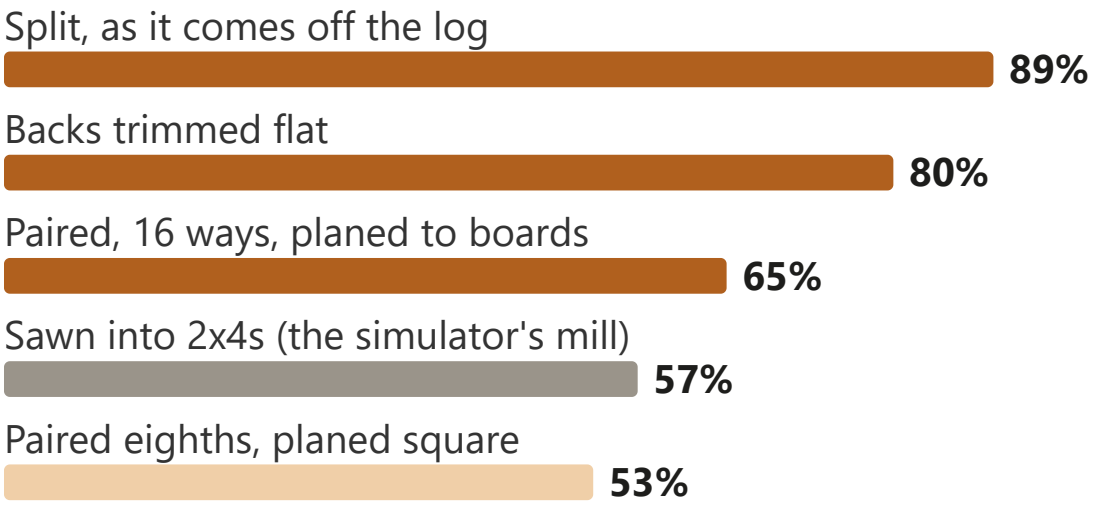
A seam is narrow. It is not a duct for a furnace, and it is not a drain for anything with solids in it. Big services go in the building's utility core, not its seams.

> **WHY THE GAP IS THE SAME EVERYWHERE**
>
> Every wedge from one split has the same face angle. Two of them side by side open a V of exactly the wedge's point angle, every time, so every seam in a wall has the same section, and one cap, one key and one fitting do all of them.

CHAPTER 10

Against the Mill

This is the chapter where the method is measured against the way everybody else does it, and where it has to say what does not help it.



How much of a 12 inch log each way of working keeps as usable wood. The grey bar is the project's sawmill simulator, packing true two-by-fours in rows across the log.

The numbers

For the reference log, 12 inches across:

- **Split eighths, as they come off the saw:** 89 per cent of the log. Only the kerf is lost.
- **The same wedges with their backs trimmed flat:** 80 per cent.
- **Split 16 ways and paired into planed boards:** 65 per cent.
- **Sawn into two-by-fours by the simulator's mill:** 57 per cent – 8 pieces a section.
- **Eighths paired and planed square:** 53 per cent.

And across the range of logs:

log, in	split, kept %	backs trimmed %	paired square %	sawn 2x4s %	2x4s a section
8	84	75	48	32	2
10	87	78	51	61	6
12	89	80	53	57	8
14	91	81	54	68	13
16	92	82	56	64	16
18	93	83	56	69	22

What these numbers say

Wedges kept as wedges beat the mill by a wide margin, at every size. That is the method's whole advantage, and it is why the dome frame comes from so few trees.

Pairs planed into boards beat the mill on the reference log. Split 16 ways, the log keeps 65 per cent as boards against the mill's 57.

Pairs planed square do not beat the mill on a good log. On logs of ten inches and more, the simulator's mill keeps more than eighths paired and planed square. Squaring every member would give the advantage away. That is the reason for the title: square only on demand, and only the members that need it.

The simulator's mill is a simple one. It packs true two-by-fours in rows and sizes each section on its small end. A good sawyer cutting mixed sizes does better; the project's books allow such a mill about 60 per cent. The wedge's lead over a good mill is smaller than its lead over the simulator, and it is still there.

Not only wood

The wedge method also skips everything a mill needs: the truck to the mill, the mill, the kiln, the truck back, the yard and the shelf. Its tools are one saw, a plane and a jig, and it is done where the tree fell. The owner timed the work: 30 wedges in a 4-hour session at the log, with the saw burning 2 tanks of fuel an hour while it rips.

> **FOR CHECKERS**
>
> The kerf, the planing allowance and the dressed lumber sizes are declared at the back of the book. Change any of them and every figure in this chapter changes with it. If you have milled logs yourself, compare your recovery with the grey bar and say which way it moved.

CHAPTER 11

Buildings That Suit Wedges

Some buildings want boards. Others are built from triangles, octagons, slopes and posts – and those are the ones wedges were made for. Here they are, with the split and the way of working that suits each.

Domes

The reference. A geodesic dome is all triangles, so its members are pushed and pulled along their length and the stick's section does not matter. The dome books frame a cabin from 120 wedges, split 8 ways, joined in pinwheels. Nothing is squared.

The wedge can fill the triangle too

Framing and infill need separate stock ledgers. In the shared nominal 72.00 inch chord example, the frame reserves 720.00 wedge feet in uncut blanks. The inward-pointing paired-infill idea then needs a measured continuous face pitch, with both wedges counted. The stated planning model gives 1,117.4 to 2,234.9 additional wedge feet; its framing inset and pair pitch are assumptions, not a solved cut list.

Changing overlap can change contact and cavities, but not the wood already in two pieces. A bowtie span is not continuous facing. The opposed-point section experiment, the tool's alternating-point flip-flop, and two points both toward the interior must not be treated as interchangeable layouts. Use the actual infill profile and stock solver before committing a woodpile. The full derivation is in the Complete Edition's Every Piece, Numbered chapter and `infill.methodology`, which the book and the film share.

Zomes

A zome is a dome made of rhombus panels in rows, a spiral rather than a sphere. Its panels can be framed flat, and the gaps where they fold are filled by wedges. It has a chapter of its own, next.

A-frames and trusses

Triangles again. Rafters can be skewed pairs, whose faces already lie at a roof pitch. Collar ties and struts stay as wedges. Only the plate the frame stands on needs to be square.

Octagons: gazebos, round decks, yurt floors

An eighth's point is the octagon's centre angle. Radial joists of trimmed wedges meet at the middle without a measured cut, and the rim is the wedges' backs.

Sheds and greenhouses

A shed roof at 3.2 in twelve is twelfths' natural slope. The walls want square posts at the corners – quarters back to back – and wedges or skewed pairs between. Greenhouse glazing bars are wedges on their backs: the glass seats on two sloping faces and the V between bars drains condensation.

Posts, fences and decks

Quarters back to back make the heaviest square in the book, with nothing to plane. Fence rails are wedges on their backs, shedding rain both ways. Deck joists are skewed pairs set on edge, decking can be sixteenths paired into boards.

Furniture

Benches, trestle tables and stools are triangles and posts. A bench seat from two skewed pairs, legs from quarters, braces from wedges.

> **THE PATTERN**
>
> Square the pieces that meet the level world: posts, plates, door and window frames. Leave everything that is triangulated, sloped or radial as wedges. Most of a building in this list is the second kind.

CHAPTER 12

Closing the Triangles

A frame of triangles is not yet a building: something has to close each triangle. Wedges can do it, and how much wood that takes is set by one choice the builder makes at the bench.

For the shared 72.00 inch nominal chord reference, the frame has 120 members: 678.35 wedge feet on the drawing, 664.31 feet of solver cut-stock, and 720.00 feet reserved as uncut blanks. These are three ledgers. Its 30 isosceles and 10 equilateral triangles leave an estimated 427.63 square feet to close for a 12.00 inch log split 8 ways. The half-chord-width frame inset is a planning assumption.

Pairs and the coverage assumption

The owner's inward-pointing pair idea is worth testing. Its timber budget requires the installed continuous face pitch, counting both wedges in each pair. The provisional model assumes pitch equals *(two minus overlap)* times the wedge chord. Under that assumption:

- the widest assumed pitch needs 1,117.4 wedge feet;
- the narrowest assumed pitch needs 2,234.9 wedge feet.

Reserve the gross frame blanks first: 90.00 trunk feet. Add the conditional infill and the total is 229.7 to 369.4 usable trunk feet, or 4 to 7 whole equivalents of a uniform 60.00 foot trunk. This excludes taper, kerf, defects, joints, seals and course offcuts.

What the dial is for

Sliding unchanged wedges changes contact and cavities, not their timber area. The ideal opposed-point chord-triangle experiment can have a half-void cell; that does not establish continuous face coverage. Opposed points, the tool's alternating-point flip-flop, and both points inward are different layouts. Neither nesting nor a cavity establishes load capacity, insulation performance or weather-tightness. Measure the chosen profile and test its joints before using the budget as a cut list. The Complete Edition's Every Piece, Numbered chapter and `infill.methodology` carry the shared derivation.

CHAPTER 13

A Zome From Wedges

A zome is built from rhombus panels set in rows that twist around a centre. The Complete Edition works one through: a zome of 36 panels in 4 rows of 9, 13.8 feet across its rim and 8.7 feet tall, every panel side 35.4 inches. This chapter builds the same zome with wedge lumber.



The Complete Edition's zome: each row is one panel, framed flat, repeated round the ring.

Panels without a bevel

Each panel is four sticks in a pinwheel, framed flat, the way a dome's triangles are. A rhombus's corners alternate sharp and blunt, and all four sticks of a panel come out the same length with the same cut at each end. No stick is bevelled. Panels meet only along their inside edges, which is where they are fastened.

The panel sticks can be skewed pairs, planed square only where they meet the next panel's face, or wedges stood on their backs, with their faces toward the inside.

The gap is a wedge

Square-edged panels that fold away from each other leave a V-shaped gap on the outside of every fold. The Complete Edition fills it with a wedge-shaped block, ripped to half the fold on each face. With wedge lumber the block is not ripped from a board: **it is a wedge, split from a log the right number of ways.** A

wedge's point is three hundred and sixty degrees divided by the split, and the gap's V is the fold, so the split each row wants is three hundred and sixty divided by its fold:

row	fold, deg	gap at depth, in	split for the block	nearest	off by, deg	by halving	off by, deg
1	24.2	1.50	14.9	15 ways	-0.2	16 ways	-1.7
2	19.3	1.19	18.7	19 ways	-0.3	16 ways	3.2
3	27.8	1.73	13.0	13 ways	-0.1	16 ways	-5.3
4	27.8	1.73	13.0	13 ways	-0.1	16 ways	-5.3

The folds run from 19.3 to 27.8 degrees, so the blocks want logs split between 13 and 19 ways. Those splits cannot be made by halving. Mark them on the end grain with a cardboard template cut to the angle, cut to the marks, and every block for that row is the same.

Halving alone gets as close as sixteenths, and the last column shows how far off that leaves each row. On a small zome that is near enough, with the difference taken up by the fastening. On a large one, cut the true split.

Why a zome suits wedges

- Its panels are flat and identical within a row, so they are built on one jig, like a dome's.
- Its folds are filled by wedges, which the log supplies at any angle you mark.
- Its rows slope, so roofing laps down it like shingles, and the V seams between panels drain the way the roof does.

> **WATCH OUT**
>
> The top rows of a zome sag under their own weight until the ring closes. Brace them during assembly, and have an engineer check the finished frame for snow and wind before anyone lives in it.

CHAPTER 14

Built by Veterans

This method was written by a veteran, and it fits the way a lot of veterans already work.

On one night in January 2024, 32,882 veterans in this country had no home of their own, and 13,851 of them were unsheltered. A building that one crew can frame in about a working week, from timber that is already standing, is one answer to that number. A village of them is a better one.

Why it fits

- **It is a system.** One body, the same parts every time, and a maintenance schedule rather than a renovation: the skin recoated on a calendar, the core serviced, a panel swapped when it wears. Anyone who has kept equipment running to a checklist already knows how to keep one of these.
- **It is crew work.** Felling, splitting, pairing and raising are jobs for a small team with a safety lead and assigned roles, and a crew can be taught the whole method in a season.
- **The skills carry.** Saw work, measuring, rigging, fabrication, repair – the trades many people learned in the service are the trades this method uses.
- **It is a place, not a unit.** Buildings on serviced pads, with a shared shop, make a community: the way a base works, without the base.

A village

The Veterans Community Project in Kansas City built a village of 49 small homes around a shared community centre. A village of wedge-built buildings could follow the same pattern: the homes, a workshop where the wedges are paired and planed, and the shop as the place where residents learn a trade and keep their own homes running.

It is designed to sit beside the programs that already exist to house veterans – HUD-VASH, SSVF, VA Grant and Per Diem – and to be built and maintained by veterans. Whether any particular project qualifies for any of them is for those programs to decide.

FOR A CREW LEADER

Run the build as a training course. One session at the log on felling and safety. One on splitting. One on pairing, shimming and planing. One on the jig. Every member of the crew does every job once, and leaves knowing how to build the next one.

CHAPTER 15

Safety at the Log

Everything in this book starts with a chainsaw in a log. That is the most dangerous part of it, and this page is the one to read twice.

- **Never alone.** Two people, minimum, at every session: one cutting, one watching, who knows how to stop the saw and call for help.
- **Wear the kit.** Chainsaw-protective chaps or trousers, a helmet with a mesh visor and ear defenders, gloves and boots. Ripping along the grain throws long strings of wood, not chips.
- **Hold the log still.** The first cut is the only one where the log can roll. Cradle it on blocks or in a pair of chocks; after the first cut, every piece has a flat face to sit on.
- **Rip with the right chain.** A chain filed for ripping cuts along the grain far faster and more safely than a crosscut chain forced to do it.
- **Keep the bar oiled and the chain sharp.** Ripping buries the bar for minutes at a time.
- **Split with the saw, always.** This method does not use axes, mauls or splitting wedges driven into the log. The saw makes every cut, along a line you marked.
- **Fasten in the thick part.** A screw near a wedge's point splits it. Drill pilot holes.
- **Have it checked.** Anything that carries people, snow or wind is checked by a licensed engineer before it is built, and members made from paired wedges are not graded lumber.

> **ONE RULE ABOVE ALL**
>
> If you are tired, stop. The owner's timed sessions at the log are 4 hours long, with good reason.

What This Book Declares

Every number in this book is computed from the geometry of a split log, except these. They are declared, each with its kind and its reason, so you can change any of them and rebuild.

declared	value	kind	why
kerf_in	0.25 in	measured	the chainsaw's cut width, the same figure every book in the project uses
plane_in	1/16 in	estimate	taken off each face when a sawn face is planed true
dressed_2x4	1.5 in	published	a dressed 2x4 is 1-1/2 by 3-1/2 inches (American Softwood Lumber Standard)
dressed_2x4_w	3.5 in	published	the 3-1/2 inch face of the same
dressed_4x4	3.5 in	published	a dressed 4x4 is 3-1/2 inches square
reference_log_in	12 in	the project's	the reference trunk the dome books are cut from

The project's measured and modelled figures – the splitting rate, the fuel, the zome, the dome's members – come from the same software as the other books in the project, and are the same numbers they print.

How this copy was made

This copy was built in one step by the project's own software: the geometry in `square_book/geometry.py`, the drawings in `square_book/figures.py`, drawn to scale from the same numbers as the tables, and the chapters in `books/square-on-demand/manuscript/`. To build another from the current manuscript:

py -3.12 -m square_book.build

It never overwrites an earlier copy: each build is a new, numbered file.